

Interview Drill

Volunteer the GAPs before they find them. Lead with strengths, own the holes.

OWN THESE 3 FIRST

1. **Mid-run crash.** In-memory state + fire-and-forget. Crash loses the run, frontend 404s. Volunteer it.
2. **How compliance scores.** They go line by line. Name the bugs yourself.
3. **Shared output dir.** Two runs clobber each other; /manifest returns the wrong run.

SAY THIS, NOT THAT

NOT "two AI agents"

SAY: two bounded orchestration modules. For-loops wrapping tool calls, no reasoning loop.

NOT "every AI call logs cost"

SAY: per-call tokens + cost for Claude. Gemini image cost is stubbed at \$0 today.

NOT "retries on the AI calls"

SAY: retries on Gemini + Dropbox. Claude prompt + vision not retried yet, known gap.

NOT "Mastra, one-import swap"

SAY: API-shape compatible so step contracts port. Real Mastra adds resume/persistence I don't have.

NOT "compliance-checked output"

SAY: two deterministic gates (color, words) + one advisory vision check. Here are the failure modes.

THE 10 QUESTIONS

1. Process restarts mid-run. What happens?

- Total loss. State only in an in-memory Map.
- Workflow runs fire-and-forget in the Express process. No queue, no checkpoint.
- Restart = empty Map. Frontend polls /status, gets 404.
- Files on disk survive but nothing points to them.

GAP: SEVERE

2. Step 4 fails on asset 30 of 100. What about 1–29?

- Whole run fails. Every step re-throws on first error.
- No partial manifest. Assets 1–29 orphaned on disk.
- Only ComplianceChecker isolates per-item.

GAP: SEVERE

3. Walk me through retry/backoff.

- $\text{base} \times 2^{(n-1)}$, +/-20% jitter, 4 attempts.
- 429/5xx retry, 4xx terminal, unknown defaults to retry.
- Applied to Gemini + Dropbox only.
- Bug to admit: terminal message hardcoded "image generation failed" even for Dropbox.

GAP: MINOR

4. Non-determinism in a compliance system?

- Color check + word scan: deterministic. Trust as gates.
- Gemini (no seed) + vision (no temp 0): non-deterministic.
- logoPresent can flip between runs. Treat vision as advisory.

GAP: MEDIUM

5. Can you reproduce a rejected image?

- No. You save the PNG + manifest, but not prompt, seed, or model version.
- Fix: capture prompt + seed + model in the manifest. Small add.

GAP: MEDIUM

6. Exactly how does compliance score? (they drill this)

- **Palette:** top-5 dominant colors, RGB distance, pass ≥ 0.3 . Loose. Scores its OWN overlay bar. Empty palette = 0 = every image FAILS.
- **Logo:** Claude vision "any logo/text?" No reference match. Campaign TEXT reads as a logo. No retry, errors swallowed to false.
- **Words:** substring scan of campaignMessage ONLY. Not the image, not OCR. False positives possible.

GAP: THE BIG ONE

7. Two campaigns POST at once. What breaks?

- Map is keyed by runId, safe. Filesystem is not.
- One shared output dir = same files overwritten.
- /manifest + /compliance read the single file, always return the LAST run.

GAP: SEVERE

8. Where's the parallelism in a 100-product run?

- None. All loops sequential, no Promise.all, no workers.
- Defensible for a POC: correctness first.
- Next: bounded per-product concurrency + a queue.

GAP: SCALE ONLY

9. uploadStatus:failed but run completes. Success or failure?

- Soft failure by design. Catches per-file, falls back to local path, run still "done".
- Reasonable, but frame it as deliberate.
- Gap: top-level status doesn't surface the partial failure.

GAP: MINOR

10. POC to production. What first?

- 1. Durable run state + job queue (runs table in the Postgres stub).
- 2. Per-run output namespacing.
- 3. Per-item isolation + partial-success manifest.
- 4. Auth + multi-tenancy (zero auth today; loadFromUrl = SSRF).
- 5. Harden compliance: temp 0, reference-logo match, OCR for words.

STRONGEST QUESTION. LEAD WITH QUEUE + RUNS TABLE.

TRUE — LEAN ON THESE

- 6 sequential steps, Zod input + output validated per step.
- 3 aspect ratios, SVG message + logo overlay.
- runId + 750ms polling (no SSE in code).
- Pluggable storage (local + dropbox live).
- Postgres deferral is honest (db is an empty stub).
- Numbers: ~\$0.12/run, ~10 AI calls, 6 creatives, image ~\$0.04 each.